

2.8b

Intelligence & Achievement

Assessing Intelligence

What is an intelligence test?

An intelligence test assesses mental aptitudes & compares them with those of others, using numerical scores.

there are 2 kinds:

Achievement tests: reflect learning, ex. AP exam

Aptitude tests: predict what you will be able to learn.

How were intelligence tests created?

Eugenicist Francis Galton wanted to measure 'natural ability' laying the racist foundation of modern intelligence testing

France, early 1900s, mandated education, but needed to measure the varied education levels

So Albert Binet & his student Théodore Simon were tasked

They assumed that all students followed same course of intellectual ability, but that some develop more rapidly.

So their test tried to measure a child's mental age.

Instead of turning to eugenics, Binet used an environmental explanation, and recommended brain exercises.

Stanford prof. Lewis Terman modified Binet's test, creating the Stanford-Binet.

From these tests, William Stern derived the infamous yet not always reliable, intelligence quotient (IQ).

$$IQ = \frac{\text{mental age}}{\text{chronological age}} \times 100$$

Original formula worked well for children, but not adults, whose rate of change of intellectual ability quickly falls.

So modern IQ tests instead represent performance relative to the average.

David Wechsler created the most widely used individual intelligence test:

Wechsler Adult Intelligence Scale.

There is one for kids, the WISC, & another for preschoolers

Are these tests accurate?

Personal

Connection

Recently, I was given the WAIS-IV test, and got a resulting FSIQ of 130

To be widely accepted a test must have psychometric properties of being:

- Standardized
- reliable
- valid



Standardization

For a test to be Standardized, test-makers give new tests to a representative sample to measure future scores against.

These scores typically form a normal curve.

The curve's highest point, the average point, is 100. Closer to extremes, fewer & fewer people.

Stanford-Binet & Wechsler scores are periodically restandardized

because of Flynn effect: the rise in int. test performance.

Reliability

A reliable test gives consistent scores when retaken

To test reliability researchers may use:

split-half: split test in equivalent halves & test each separately

test-retest: retest with same test

Stanford-Binet, WAIS & WISC are very reliable after childhood

Validity

Validity is how much a test measures what it's supposed to

tests that accurately measure behavior have content validity

tests that accurately measure concepts have construct validity

tests that predict future performance have predictive validity

Predictive power of aptitude tests peaks in early school years.

SAT is weaker at predicting uni grades

GRE is even weaker predicting grad school performance

Predicting power gets worse because the range gets smaller in higher ed.

MCQ

1. d 2. c 3. a 4. d 5. c 6. a

7. c 8. b